

1. Project Overview

Objective

Develop a **Transportation Management System (TMS)** to manage:

- Transport requests
 - Dispatching
 - Waybill generation
 - Fleet (own + outsourced)
 - POD tracking
 - Operational visibility
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2. System Modules (Core Architecture)

1. Client Management

- Client Master Data
 - Contract rates (Depends on destinations/Truck Type)
 - Contact persons
 - Client's pick up locations
 - Client's delivery locations
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2. Transport Request Module

User creates request with:

Required Fields:

- Client Name (dropdown +Search Option)
- Pickup Location (dropdown)
- Delivery Location (dropdown)
- Shipment Type (Dry / Cold / Frozen)
- Truck Type (Flatbed / Curtain / 5T / 10T / Van)
- Quantity / Weight / Pallets
- Required Pickup Date & Time
- Delivery Date
- Special Instructions
- PO Number / Reference

Status:

- Draft
 - Submitted
 - Approved
 - Rejected
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3. Dispatching Module

Dispatch team receives requests and:

- View all open requests (dashboard)
 - Assign:
 - Truck (Own / Outsource)
 - Driver
 - Supplier (if outsourced)
 - Update status:
 - Assigned
 - In Transit
 - Delivered
 - Issue
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4. Fleet Management

A. Own Fleet (dropdown)

- Truck ID
 - Plate Number
 - Type
 - Capacity
 - Driver Assigned
 - Status (Available / Busy / Maintenance)
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B. Outsourced Suppliers (dropdown)

- Supplier Name
 - Contact details
 - Truck types available
 - Rate card (optional)
 - Performance tracking (future)
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5. Waybill Module

Auto-generated after dispatch

Contains:

- Waybill Number (Auto)
- Client Details
- Pickup / Delivery (Google Maps Licks)
- Shipment details
- Assigned truck / driver
- Supplier (if outsourced)

Output:

- PDF Waybill
 - Printable format
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6. Communication Module

Options:

- Send Waybill via:
 - Email
 - WhatsApp (If possible)
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7. Shipment Tracking

Status flow:

- Assigned
- Picked Up
- In Transit
- Delivered

Manual updates by:

- Dispatch team
 - Driver (optional mobile)
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8. POD (Proof of Delivery)

Upload options:

- Mobile upload
- Office upload

Attach:

- Signed delivery note
- Photos

Linked to:

- Waybill
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9. Exception Management

Capture:

- Delays
 - Damages
 - Customer complaints
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10. Reporting Dashboard

Key metrics:

- Active shipments
 - Completed shipments
 - Delays
 - Fleet utilization
 - Own vs outsourced ratio
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3. End-to-End Workflow

Step 1: Create Request

Client / Internal team:
→ Creates Transport Request

Step 2: Approval (Optional)

Manager:
→ Approves request

Step 3: Dispatch

Dispatch team:
→ Assigns:

- Truck
- Driver
- Supplier (if needed)

Step 4: Waybill Generation

System:
→ Auto-creates Waybill
→ Generates PDF

Step 5: Send Waybill

→ Sent via:

- WhatsApp
- Email

Step 6: Shipment Execution

Status updates:

- Picked up
 - In transit
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Step 7: Delivery

→ Mark as Delivered

Step 8: POD Upload

→ Upload signed document & Photos of Loading & Unloading

Step 9: Close Shipment

→ Ready for billing

4. User Roles

1. Admin

- Full access

2. Operations

- Create requests
- Track shipments

3. Dispatch

- Assign trucks
- Manage execution

4. Driver (Optional Mobile)

- Update status
- Upload POD
- Upload Photos

5. Finance (Future)

- Invoice tracking (Zoho Books)
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5. Key Features (Critical)

- Unique Waybill Number
 - Status tracking (real-time)
 - PDF generation
 - Mobile access
 - Role-based access
 - Search & filtering
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6. Integrations (Phase 2)

- WhatsApp API
 - Zoho Books (invoicing)
 - GPS tracking
 - SMS alerts
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7. Important Notes for Developer

- System must handle:
 - Own fleet AND outsourced fleet
 - Must be scalable
 - Must support mobile usage
 - Must allow future integrations
 - **“We are not building a basic request system.
We are building a simplified TMS that can scale with operations.”**
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